

소규모 집단의 한계를 넘어:

제주흑우 후보 씨수소 조기평가를 위한 단계적 다중정보 기반 AI 방법론

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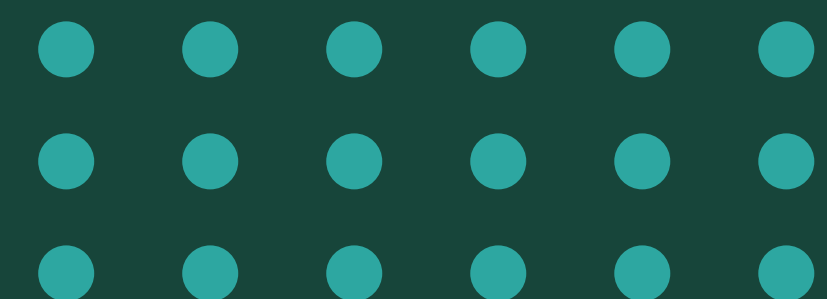
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THE SMALL-POPULATION CHALLENGE

How can future sire candidates be evaluated early when both animals and reference records are limited?

LARGE-POPULATION SETTING



More candidates enable stronger selection

Conceptual comparison - not animal counts

JEJU BLACK CATTLE REALITY



Few candidates: characterize each bull comprehensively

Pilot: repeated, age-aligned evaluation of all eligible bulls



Restricted candidate pool

Limited reference records¹

Greater relatedness

Overfitting / unstable rankings

Large-population selection cannot simply be scaled down.

DESIGN PRINCIPLE

Extract richer longitudinal evidence from each bull without treating repeated records as additional animals

REPEATED
AGE-ALIGNED
MULTIMODAL

A STEPWISE METHOD DESIGNED FOR SMALL n : Conceptual framework - evaluate added information before adding complexity

01 CAPTURE DEVELOPMENT

The same bull is repeatedly measured within prespecified age windows



QC → AGE ALIGNMENT → LEVEL • SLOPE • CHANGE

Body weight & growth rate

RGB-D/3D morphometrics + BCS²

Feeding records

Metabolic & endocrine markers

Early-life scrotal/testicular development

TRAJECTORY > SINGLE SNAPSHOT⁵

02 TEST INFORMATION VALUE

Compare a longitudinal mixed-effects benchmark with parsimonious ML

M0 Age-aligned BW/growth trajectory

M1 + RGB-D/3D morphometrics + BCS

M2 + feeding, metabolic and endocrine markers

M3 + early reproductive development

Retain an information layer only when improvement is stable across outer validation folds

More variables ≠ a better model

Benchmark: mixed-effects growth model |

Candidate ML: elastic net / PLS

03 VALIDATE HONESTLY

Repeated records and closely related bulls do not provide independent evidence

PRIMARY TARGET + AGE CUTOFF

Predefine the primary later-life breeding-soundness or semen endpoint, secondary outcomes, and cutoff age; use only earlier measurements as predictors

ANIMAL-BLOCKED CV³ + KINSHIP SENSITIVITY

Keep all records from each bull in one validation fold; group close relatives in a sensitivity analysis where feasible

NO RECORD- OR PIPELINE-LEVEL LEAKAGE

Perform preprocessing, feature selection and model tuning within training folds only

REPORT: prediction error • uncertainty • rank stability

04 PRIORITIZE, THEN CONFIRM: The model prioritizes candidates for confirmatory testing; it does not make the final breeding decision

ALL ELIGIBLE
YOUNG BULLS

EARLY-LIFE PRIORITY
RANKING + UNCERTAINTY

CONFIRMATORY EVALUATION
Breeding soundness + semen testing
Genomic relationship + inbreeding review⁴

SUPPORT FINAL
SIRE SELECTION

EVALUATION CRITERIA

ADDED VALUE: Does added data improve prediction beyond body weight?

ROBUSTNESS: Consistent across folds and age cutoffs?

EARLY VALIDITY: Do early rankings agree with later outcomes?

RELIABILITY: Are errors, uncertainty and rankings stable?

PRACTICAL VALUE

EARLIER: Prioritize promising young bulls sooner

EFFICIENT: Focus costly tests on priority candidates

DIVERSITY-AWARE: Consider genomic relationships and inbreeding

TAKE-HOME Age-aligned multimodal trajectories support early sire evaluation without replacing final breeding decisions

¹ Haque MA, et al. *Anim Biosci.* 2026;39(8):260150. doi:10.5713/ab.260150

² Hanson EM, et al. *Front Anim Sci.* 2026;7:1752658. doi:10.3389/fanim.2026.1752658

³ Coelho Ribeiro LA, et al. *J Anim Sci.* 2021;99:skab206. doi:10.1093/jas/skab206

⁴ Bem RD, et al. *BMC Genomics.* 2024;25:738. doi:10.1186/s12864-024-10641-3

⁵ Shokrollahi B, et al. *Biology.* 2026;15:1522. doi:10.3390/biology15171522